Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Intelligent Analysis of Engineering Documentation Using AI			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Knowledge Services		Knowledge Services	
	Medium Category	Small Category	Medium Category	Small Category
	R&D / Engineering Services	Production Management / Quantitative Analysis Techniques	Management Strategy / Finance / Trade Services	Service Networks / Collaboration Support
Concept of Proposed Technology	The proposed technology leverages advanced artificial intelligence (AI) models to intelligently analyze and interpret engineering documentation, including technical drawings and textual documents. This solution aims to overcome the challenges posed by the complexity, variability, and volume of engineering data by automating the recognition and extraction of critical information. Key components of the technology include: 1. **AI-Driven Interpretation**: Utilizing Computer Vision techniques for technical drawing recognition and Large Language Models (LLMs) for understanding textual documents. 2. **Automated Data Structuring**: Transforming unstructured engineering data into structured formats for easier comprehension and utilization. 3. **Scalability and Adaptability**: Supporting diverse industries and standards by implementing scalable and customizable AI frameworks. This technology addresses the inefficiencies in manual analysis of engineering documentation, such as time consumption, human error, and limited scalability. By automating these processes, it enhances productivity, ensures accuracy, and facilitates seamless integration into existing workflows. Expected outcomes include improved engineering precision, reduced operational costs, and the			

	ability to handle large-scale documentation efficiently. The proposed solution paves the way for smarter, faster, and more reliable engineering practices, supporting innovation and competitiveness in the industry.
Rationale for Joint R&D Project	**For Korean SMEs**: Access to German research expertise enhances their technological capabilities, allowing them to scale their solutions globally and improve competitiveness in the market. **For German Research Institutions**: Collaborating with Korean SMEs to transform research outcomes into practical application technologies, create impactful solutions through Korea's rapid technology commercialization abilities. By combining these strengths, the project aims to create technology that is both globally interoperable and scalable, ensuring its adoption across various industries and markets worldwide.
Key Words	1. Document Intelligence 2. AI 3. LLM 4. Automation 5. Digitalization

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	- Develop advanced AI models to interpret technical documents provided by the industry partner - Automate the extraction and structuring of data to streamline the processing of the documents - Enhance interoperability across industries by ensuring compliance with international standards - Integrate recognized information into the industry partner's existing systems			
R&D Target Specifications	**Accuracy in document recognition**: Achieve an interpretation accuracy of at least 90% for technical documents, including both diagrams and textual documents. **Processing speed**: Enable the system to process and extract data from 20 pages of engineering documents per minute, ensuring scalability for large-scale applications. **Interoperability**: Ensure usability for every drawing type corresponding to one of 80% or more of major international engineering standards (e.g., ISO, DIN). **Reduction in analysis time**: Reduce the time required for workers to analyze technical documentation by at least 50%, improving overall operational efficiency. *(Note: The accuracy and processing speed may vary depending on the complexity of the partner's engineering documents.)*			
R&D Plan				

Work Package	Research approach, Key task, Methodology	Project Month
Phase I (Pre-study): 6 months		
WP1	Definition of company specific requirements: To ensure a target development of the technical drawing interpretation the customer specific requirements and needs will be collaboratively defined in common workshops. At the same time, the generalizability and scalability of the solution will be still maintained.	PM1 – PM2
WP2	Data exploration and collection: After the requirements definition (WP1), the data will be explored to assess critical quality aspects like completeness and redundancy. If necessary, data processing steps including labeling and data cleaning will be carried out.	PM2 – PM3
WP3	First tryout: Based on the requirements (WP1) and the prepared data (WP2) the first tryout will be conducted with pre-developed tools (LLM, Computer Vision). Minor fine-tuning will be performed if needed. The results of this first trial will provide a clear assessment of the feasibility of the approach and help identify necessary tasks for further development towards the productization stage.	PM3 – PM6
Phase II (Main R&D): 30 months		
WP4	Advanced tools for technical drawing interpretation: In WP3 new functions of the technical drawing interpretation tool have been defined. WP4 will focus on developing full-scale AI models, including advanced LLM for textual analysis and computer vision systems for drawing recognition, to implement these defined functions and deliver a complete solution for the industry partner. During this development phase, algorithms will be optimized to enhance accuracy and efficiency. Intermediate results will be continuously validated to ensure alignment with the defined requirements and objectives. Additionally, all specific standards will be considered to guarantee both compliance and interoperability.	PM7 – PM18

WP5	Workflow integration and testing: The newly developed tools will be integrated into the daily workflow of the industry partner, utilizing either an on-prem or cloud-based solution approach. The integrated software will undergo validation under live operating conditions and be evaluated against the defined Key Performance Indicators (refer to R&D target specifications). Feedback gathered from the day-to-day use of the drawing interpretation tools will help to further optimizing and fine-tuning the solution to ensure maximum efficiency and usability.	PM19 – PM30
WP6	Generalization and Scaling: Following the successful implementation tailored to the specific needs of the industry partner, the developed solution will be further generalized to ensure scalability and applicability across a broader range of global companies. To achieve this, additional relevant industrial standards, such as ISO and DIN, will be considered to enhance compatibility and facilitate widespread adoption.	PM31 – PM36

Milestones:
M1 (PM6): Feasibility is assessed and tasks for further development towards the productization are defined
M2 (PM18): Advanced tools for technical drawing interpretation are developed and made available
M3 (PM30): Drawing interpretation software is successfully tested and validated in an industrial environment
M4 (PM36): Generalized and scalable solution is ready for adoption by industrial global companies

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- Develop advanced AI models, including large language models (LLMs), computer vision systems, tailored for the interpretation of complicated technical documents. - Conduct research on innovative algorithms to improve the accuracy and efficiency of document recognition and data extraction processes - Deploy the developed models within the company's current infrastructure - Provide technical guidance and expertise to ensure project's outcomes
Expected R&R of Korean SME Partner	- Provide domain expertise in industrial processes and engineering workflows - Identify specific requirements and challenges in analyzing technical documents within their industry - Share necessary data for training and validation for AI models - Implement supportive software systems and platforms for integrating AI solutions into existing workflows - Conduct comprehensive testing of the developed AI models using real-world engineering datasets - Provide feedback and insights to refine model performance and ensure practical applicability

5. Expected Outcomes/Impact

Expected Outcomes and Impact	1. Technological Achievements • Creation of scalable and adaptable AI models that can process large volumes of complex engineering data with high accuracy and efficiency. • Integration of innovative algorithms into industry workflows, ensuring seamless interoperability with existing systems and international standards. 2. Commercialization Potential • Deployment of the developed technology into the industry partner's operations, enabling faster and more accurate processing of engineering documentation. • Expansion into global markets by leveraging the technology's compliance with major international engineering standards. 3. Broader Industrial Impact • Reduction in time and costs associated with manual analysis of engineering documents, improving operational efficiency and productivity. • Facilitation of digital transformation within industries, promoting the adoption of automation and AI in documentation workflows. 4. Economic Impact • Enhanced competitiveness of participating SMEs in the global market through access to cutting-edge AI technology, by establishing a model for successful international R&D collaboration that inspires future partnerships between Korea and Germany.

6. References

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(3) "DeepPatent2: A Large-Scale Benchmarking Corpus for Technical Drawing Understanding", K. Ajayi, X. Wei, M. Gryder, W. Shields, J. Wu, S. M. Jones, M. Kucer, D. Oyen

(4) "Development of an engineering drawing detection and extraction algorithm for quality inspection using deep neural networks", M. M. Agritania, M. M. Isnaini, CIRPe 2024

(5) "Extraction of dimension requirements from engineering drawings forsupporting quality control in production processes", B. Scheibel, J. Mangler, S. Rinderle-Ma, Computers in Industry 2021

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization